Grid-Scale Energy Storage Systems

Market Outlook 2025–2030

A Grid-Scale Energy Storage Systems proosures innouediable firm-consucting and enermy, nnnatal, and energes energies.

PUBLICATION DATE	REPORT LENGTH	REPORT ID
November 2025	38 Pages	HH-ER-2025-047

www.hheuristics.com

© Copyright 2022 hheuristicssulting firm. All rights reserved.

Table of Contents

1. Executive Summary 3

2. Market Size & Growth Forecast 5

3. Technology Analysis 12

4. Competitive Landscape 18

5. Market Dynamics & Business Models 22

6. Regional Deep Dives 26

7. Regulatory & Policy Environment 30

8. Strategic Recommendations 34

9. Methodology & Sources 37

EXHIBITS

Exhibit 1: Global Grid-Scale Storage Capacity & Revenue (2020–2030F) 6

Exhibit 2: Market Revenue by Technology Segment 8

Exhibit 3: Annual Deployment by Region (GW) 10

Exhibit 4: Battery Technology Comparison Matrix 12

Exhibit 5: Lithium-ion Battery Pack Cost Decline 14

Exhibit 6: Competitive Market Share Analysis (2024) 19

Exhibit 7: Revenue Stream Analysis 22
Exhibit 8: Revenue Stack by Market 24
Exhibit 9: Investment Decision Framework 34

01.
Executive Summary

Key findings and strategic implications from our analysis of the global grid-scale energy storage market.

The global grid-scale energy storage market is entering a transformative period of mainstream adoption, driven by accelerating renewable energy deployment, declining battery costs, and increasingly supportive policy frameworks. This report provides a comprehensive analysis of the market's trajectory through 2030.

- **Market Scale:** The global grid-scale storage market reached an estimated $15.8 billion in revenue in 2024, with cumulative installed capacity surpassing 110 GW. We project the market to reach $47.2 billion by 2030, representing a compound annual growth rate (CAGR) of 20.1%.
- **Technology Leadership:** Lithium iron phosphate (LFP) chemistry has emerged as the dominant technology for utility-scale applications, capturing 70% of market revenue in 2024. LFP's superior safety profile, longer cycle life (6,000+ cycles), and lower cost have driven this transition.
- **Regional Dynamics:** China leads global deployment with 45% of annual installations, followed by the United States (28%) and Europe (15%). The U.S. market has accelerated dramatically following the Inflation Reduction Act (IRA).
- **Emerging Technologies:** Sodium-ion batteries entered commercial deployment in 2024 and are projected to capture 8–12% of the grid-scale market by 2030. Long-duration energy storage (LDES) technologies are attracting significant investment.

- **Competitive Landscape:** CATL maintains market leadership with 34% share, followed by BYD (18%) and Tesla (13%). Software capabilities have emerged as a critical source of competitive differentiation.

KEY FINDING

Battery pack costs have declined approximately 90% since 2010 — from over $1,100/kWh to ~$139/kWh in 2024 — making storage economically competitive with conventional peaking generation across an expanding range of applications and geographies.

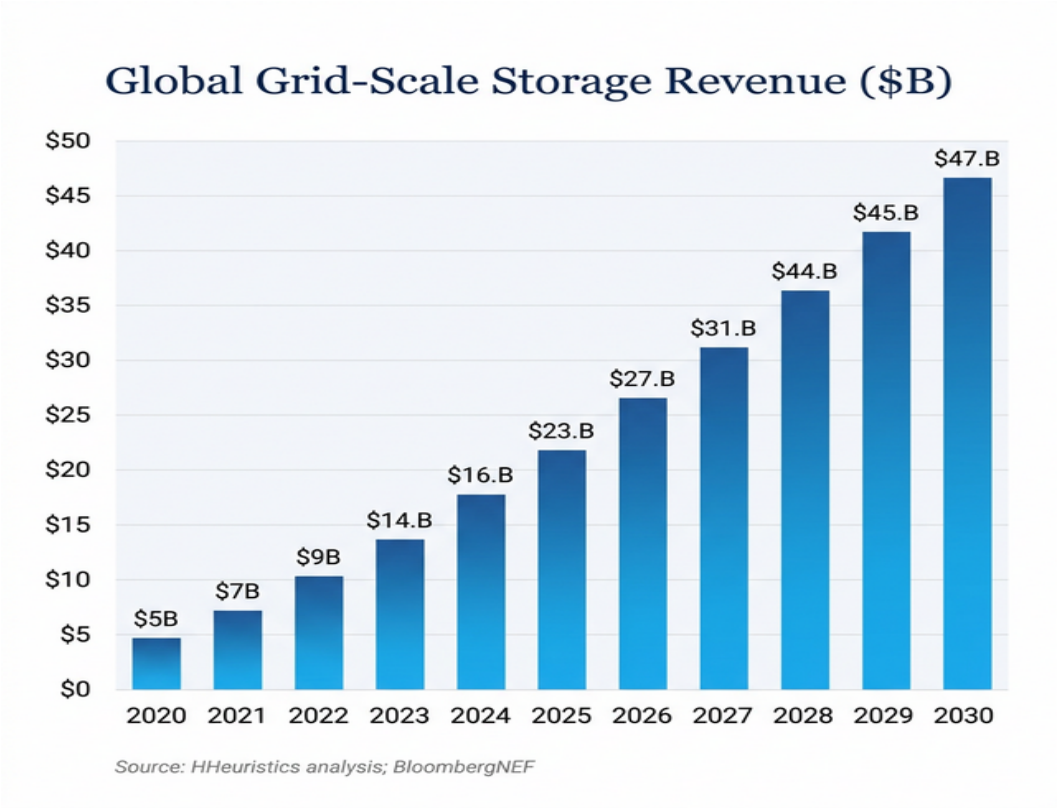
02.
Market Size & Growth Forecast

Global market overview, technology segmentation, and regional deployment analysis.

2.1 Global Market Overview

The global grid-scale energy storage market has experienced exceptional growth over the past five years, with installed capacity expanding from approximately 17 GW in 2020 to over 85 GW by end of 2024 — a compound annual growth rate of 49.6% in capacity terms. Market revenues have grown commensurately, rising from $4.8 billion in 2020 to an estimated $15.8 billion in 2024.

EXHIBIT 1: GLOBAL GRID-SCALE STORAGE REVENUE



Source: HHeuristics analysis; BloombergNEF; Wood Mackenzie

EXHIBIT 1: GLOBAL GRID-SCALE STORAGE CAPACITY & REVENUE (2020–2030F)

YEAR	CUMULATIVE CAPACITY (GW)	ANNUAL ADDITIONS (GW)	MARKET REVENUE ($B)	YoY GROWTH
2020	17.4	4.7	$4.8	—
2021	27.2	9.8	$6.9	44%
2022	43.4	16.2	$9.4	36%
2023	68.7	25.3	$12.1	29%
2024E	110.5	41.8	$15.8	31%
2025F	163.2	52.7	$20.2	28%
2026F	225.8	62.6	$25.4	26%
2027F	299.1	73.3	$31.2	23%
2028F	383.7	84.6	$37.8	21%
2029F	478.4	94.7	$42.4	12%

YEAR	CUMULATIVE CAPACITY (GW)	ANNUAL ADDITIONS (GW)	MARKET REVENUE ($B)	YoY GROWTH
2030F	585.0	106.6	$47.2	11%

Source: HHeuristics analysis; BloombergNEF; Wood Mackenzie; company disclosures

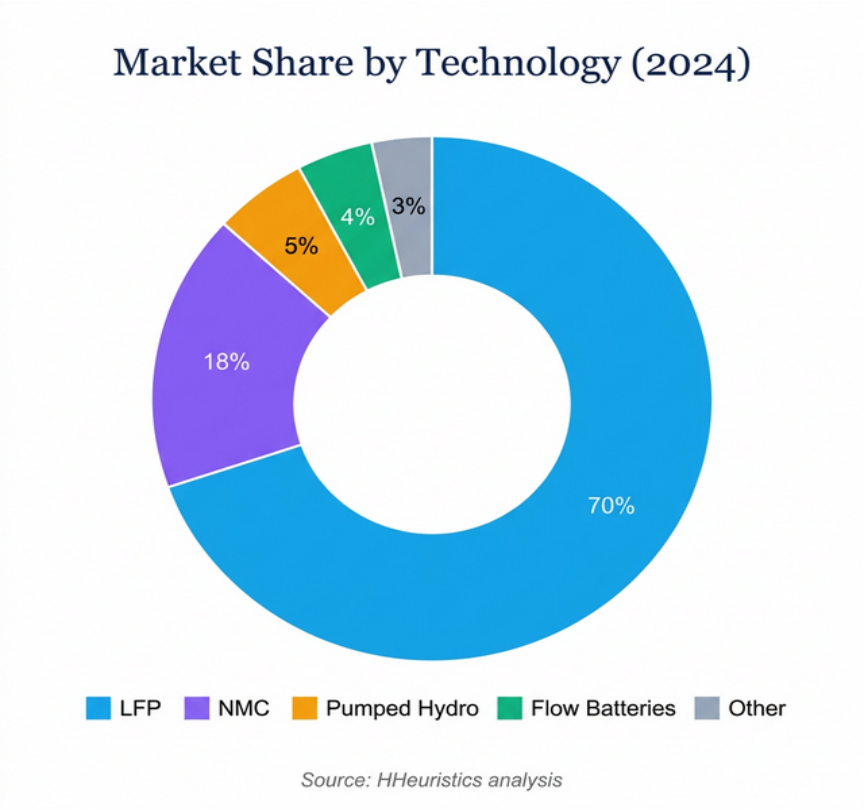
FORECAST SCENARIOS

- **Bull Case ($58B by 2030):** Accelerated policy support, faster cost declines, breakthrough in LDES technologies
- **Base Case ($47B by 2030):** Continued policy support, gradual cost reductions, moderate supply chain constraints
- **Bear Case ($38B by 2030):** Policy reversals, persistent supply chain disruptions, slower renewable deployment

2.2 Market Segmentation by Technology

Lithium-ion technology continues to dominate the grid-scale storage market, accounting for 92% of new capacity additions in 2024. However, the technology mix is evolving as sodium-ion and long-duration storage technologies gain traction.

EXHIBIT 2: MARKET SHARE BY TECHNOLOGY (2024)



Source: HHeuristics analysis

EXHIBIT 2: MARKET REVENUE BY TECHNOLOGY SEGMENT (2024 VS. 2030F)

TECHNOLOGY	2024 REVENUE	2024 SHARE	2030F REVENUE	2030F SHARE	CAGR
Lithium-ion (LFP)	$11.1B	70%	$28.3B	60%	16.9%
Lithium-ion (NMC)	$2.8B	18%	$5.2B	11%	10.8%
Sodium-ion	$0.3B	2%	$4.2B	9%	54.8%
Flow Batteries	$0.6B	4%	$3.3B	7%	32.7%
Other LDES	$0.2B	1%	$2.8B	6%	55.2%
Pumped Hydro	$0.8B	5%	$3.4B	7%	27.3%
Total	**$15.8B**	**100%**	**$47.2B**	**100%**	**20.1%**

Source: HHeuristics analysis; BloombergNEF

TECHNOLOGY OUTLOOK

> Sodium-ion batteries are projected to capture 8–12% of the grid-scale storage market by 2030, primarily displacing LFP in cost-sensitive applications. The technology is particularly attractive in markets seeking to reduce dependence on lithium supply chains.

03.
Technology Analysis

Comprehensive assessment of current and emerging storage technologies.

3.1 Technology Overview

Grid-scale energy storage encompasses a diverse range of technologies, each with distinct performance characteristics, cost structures, and optimal applications.

EXHIBIT 4: BATTERY TECHNOLOGY COMPARISON MATRIX

PARAMETER	Li-ion (LFP)	Li-ion (NMC)	SODIUM-ION	VANADIUM FLOW	IRON-AIR
Energy Density (Wh/L)	250–350	400–600	150–250	25–35	750–1,000
Cycle Life	4,000–8,000	2,000–4,000	3,000–6,000	15,000–20,000	3,000–5,000
Round-trip Efficiency	92–96%	90–94%	88–92%	70–80%	45–55%
Duration Sweet Spot	2–4 hrs	2–4 hrs	2–6 hrs	4–12 hrs	24–100 hrs
Capital Cost ($/kWh)	$150–200	$180–250	$100–150	$350–500	$50–100 (target)
Commercial Maturity	Mature	Mature	Early commercial	Commercial	Demo
Safety Profile	Good	Moderate	Excellent	Excellent	Excellent

Source: HHeuristics analysis; BNEF; NREL; manufacturer specifications

3.2 Lithium-Ion Technologies

Lithium Iron Phosphate (LFP)

LFP chemistry has become the dominant technology for grid-scale storage applications, capturing 78% of utility-scale deployments in 2024. Key advantages include:

- **Superior safety:** Inherently stable chemistry with minimal thermal runaway risk
- **Longer cycle life:** 6,000+ cycles at 80% depth of discharge, enabling 15–20 year project lifetimes
- **Lower cost:** Freedom from cobalt and nickel supply constraints; costs approaching $130/kWh at pack level
- **Supply chain security:** Mature manufacturing base, primarily in China

EXHIBIT 5: LITHIUM-ION BATTERY PACK COST DECLINE ($/KWH)

YEAR	LFP PACK PRICE	NMC PACK PRICE	LFP YoY CHANGE
2018	$176	$185	—
2019	$156	$161	-11%
2020	$137	$140	-12%
2021	$132	$138	-4%
2022	$153	$165	+16%
2023	$130	$149	-15%
2024E	$115	$139	-12%
2026F	$98	$118	—
2028F	$89	$108	—
2030F	$82	$98	—

Source: BloombergNEF; BNEF Battery Price Survey; HHeuristics analysis

04.
Competitive Landscape

Market structure, leading competitors, and competitive dynamics.

4.1 Market Structure

The grid-scale energy storage market features a complex competitive landscape with participants across multiple segments of the value chain. The market exhibits increasing vertical integration, with leading cell manufacturers expanding into system integration and software capabilities emerging as a critical source of competitive differentiation.

EXHIBIT 6: COMPETITIVE MARKET SHARE ANALYSIS (2024)

COMPANY	HQ	PRIMARY POSITION	2024 REV ($B)	MARKET SHARE
CATL	China	Cell Manufacturer	$5.4	34%
BYD	China	Integrated	$2.8	18%
Tesla	United States	System Integrator	$2.1	13%
Fluence	United States	System Integrator	$1.4	9%
Samsung SDI	South Korea	Cell Manufacturer	$0.8	5%
LG Energy Sol.	South Korea	Cell Manufacturer	$0.7	4%
Sungrow	China	Integrated	$0.9	6%
Others	Various	Various	$1.7	11%
Total			**$15.8**	**100%**

Source: Company filings; HHeuristics analysis

08.
Strategic Recommendations

Investment framework and market entry prioritization for key stakeholders.

8.1 Market Entry Prioritization

For organizations considering geographic expansion, HHeuristics recommends prioritizing markets based on the following criteria:

PRIORITY	MARKET	RATIONALE
1 (Highest)	United States (ERCOT, CAISO)	IRA incentives; large pipeline; sophisticated markets
2	United Kingdom	Mature market; strong revenue opportunities; clear policy
3	Australia	High renewables; price volatility; supportive state policies
4	Germany / Italy	Growing markets; EU policy support; grid needs
5	India	High growth potential; scale opportunity; execution risk

8.2 Risk Factors

- **Policy risk:** Changes to incentive programs or market rules
- **Supply chain disruption:** Raw material shortages; geopolitical tensions
- **Technology risk:** Faster-than-expected obsolescence; safety incidents

- **Market saturation:** Oversupply in certain markets reducing revenue
- **Execution risk:** Interconnection delays; cost overruns; operational issues

09.
Methodology & Sources

Research methodology, data sources, and limitations.

9.1 Research Methodology

This report was developed using HHeuristics' proprietary research methodology, integrating multiple data sources and analytical approaches.

Primary Research

- In-depth interviews with 18 industry executives spanning manufacturers, developers, utilities, and investors
- Survey of 45 project developers on market conditions and investment priorities
- Site visits to manufacturing facilities and operational storage projects

Data Sources

- BloombergNEF Energy Storage Outlook and Battery Price Survey
- Wood Mackenzie Global Energy Storage Service
- International Energy Agency (IEA) World Energy Outlook
- U.S. Energy Information Administration (EIA)
- National laboratory reports (NREL, Sandia, PNNL)

Disclaimer: This report is provided for informational purposes only and does not constitute investment advice. HHeuristics makes no representations or warranties regarding the accuracy or completeness of the information contained herein. Readers should conduct their own due diligence before making investment decisions. Past performance is not indicative of future results. © 2025 HHeuristics. All rights reserved. No part of this publication may be reproduced without prior written permission.

9.4 About HHeuristics

HHeuristics provides in-depth, data-driven market research reports across a wide range of industries, offering actionable insights for investors, executives, policymakers, and analysts.

Website: www.hheuristics.com

Email: research@hheuristics.com

Reports: Available at MarketResearch.com